School Worksheet

Class 12 Mathematics, Chapter 7: Integrals

Maximum marks: **54** **20** questions

Name _____ Date _____ Score _____ / 54

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

$\int e^x(\sin x + \cos x)\,dx$ is equal to:

(A) $e^x \cos x + C$

(B) $e^x \sin x + C$

(C) $-e^x \sin x + C$

(D) $e^x(\sin x - \cos x) + C$

2 MCQ — 1 mark

If $\int \frac{2^{1/x}}{x^2}\,dx = k \cdot 2^{1/x} + C$, then k is equal to:

(A) $-\frac{1}{\log 2}$

(B) $-\log 2$

(C) -1

(D) $\frac{1}{2}$

3 MCQ — 1 mark

The value of $\int_1^e \log x\,dx$ is:

(A) 0

(B) 1

(C) e

(D) $e \log e$

4 MCQ — 1 mark

$\int 2^{x+2}\,dx$ is equal to:

(A) $2^{x+2} + C$

(B) $2^{x+2} \log 2 + C$

(C) $\frac{2^{x+2}}{\log 2} + C$

(D) $2 \cdot \frac{2^x}{\log 2} + C$

Q No.	Question	Marks
5	MCQ If $\int_0^1 \frac{e^x}{1+x}\,dx = \alpha$, then $\int_0^1 \frac{e^x}{(1+x)^2}\,dx$ is equal to: **(A)** $\alpha - 1 + \frac{e}{2}$ **(B)** $\alpha + 1 - \frac{e}{2}$ **(C)** $\alpha - 1 - \frac{e}{2}$ **(D)** $\alpha + 1 + \frac{e}{2}$	1 mark
6	Assertion-Reason **Assertion (A):** $\int_{-1}^{1} x\|x\|\,dx = 0$. **Reason (R):** $f(x) = x\|x\|$ is an odd function, and $\int_{-a}^{a} f(x)\,dx = 0$ for every odd function f. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Find $\int \frac{x^3 - x^2 + x - 1}{x - 1}\,dx$.	2 marks

Level 2

19 marks

8	Short answer Evaluate: $\int_0^{\pi/2} \sin 2x \cos 3x\,dx$.	2 marks
9	Short answer Given $\frac{d}{dx}F(x) = \frac{1}{\sqrt{2x - x^2}}$ and $F(1) = 0$, find $F(x)$.	2 marks
10	Short answer Find: $\int \frac{2x - 1}{(x-1)(x+2)(x-3)}\,dx$.	3 marks
11	Short answer Evaluate $\int_0^{\pi/4} \log(1 + \tan x)\,dx$.	3 marks
12	Short answer Find $\int e^x \left(\frac{1 + \sin x}{1 + \cos x}\right) dx$.	3 marks

Q No.	Question	Marks
13	Short answer Evaluate: $\int_0^5 (\lvert x-1\rvert + \lvert x-2\rvert + \lvert x-5\rvert)\,dx.$	3 marks
14	Short answer Find $\int \sin^3 x \cos^2 x\,dx.$	3 marks

Level 3 — 27 marks

15	Long answer Evaluate $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\,dx.$	5 marks
16	Long answer Find $\int \frac{x^2+1}{(x-1)^2(x+3)}\,dx.$	5 marks
17	Long answer Evaluate $\int_0^{\pi/2} \log \sin x\,dx.$	5 marks
18	Short answer Find $\int \frac{x+2}{\sqrt{x^2+2x+3}}\,dx.$	3 marks
19	Long answer Find $\int \left(\sqrt{\tan x} + \sqrt{\cot x}\right) dx.$	5 marks
20	Case study **Case Study:** In class, a teacher demonstrates the property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, which makes many definite integrals easy to evaluate. **(i)** Evaluate $\int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x}\,dx.$ — 1 mark **(ii)** Evaluate $\int_0^{\pi/2} \frac{\sin^n x}{\sin^n x + \cos^n x}\,dx.$ — 1 mark **(iii)** Evaluate $\int_0^1 x(1-x)^{99}\,dx.$ — 2 marks	4 marks

Answer key

Chapter 7: Integrals, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) $e^x \sin x + C$
2	(A) $-\dfrac{1}{\log 2}$
3	(B) 1
4	(C) $\dfrac{2^{x+2}}{\log 2} + C$
5	(B) $\alpha + 1 - \dfrac{e}{2}$
6	(A) Both true; R is the correct explanation of A
7	$\dfrac{x^3}{3} + x + C$

Level 2

8	$-\dfrac{2}{5}$
9	$F(x) = \sin^{-1}(x-1)$
10	$-\dfrac{1}{6}\log\lvert x-1\rvert - \dfrac{1}{3}\log\lvert x+2\rvert + \dfrac{1}{2}\log\lvert x-3\rvert + K$
11	$\dfrac{\pi}{8}\log 2$
12	$e^x \tan\dfrac{x}{2} + C$
13	$\dfrac{55}{2}$
14	$-\dfrac{\cos^3 x}{3} + \dfrac{\cos^5 x}{5} + C$

Level 3

15	$\dfrac{\pi^2}{4}$
16	$\dfrac{3}{8}\log\lvert x-1\rvert - \dfrac{1}{2(x-1)} + \dfrac{5}{8}\log\lvert x+3\rvert + K$
17	$-\dfrac{\pi}{2}\log 2$
18	$\sqrt{x^2+2x+3} + \log\left\lvert x+1+\sqrt{x^2+2x+3}\right\rvert + C$
19	$\sqrt{2}\,\sin^{-1}(\sin x - \cos x) + C$
20	(i) $\dfrac{\pi}{4}$ (ii) $\dfrac{\pi}{4}$ (iii) $\dfrac{1}{10100}$